



Dualism and Structural Transformation: The Informal Manufacturing Sector in India

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Abstract

We identify a basic dualism within the informal manufacturing sector (IMS) in India between a ‘traditional’/non-capitalist segment, comprising family-based household enterprises that constitute the vast majority of the IMS, and a segment of ‘modern’/capitalist enterprises employing wage labour. We focus on the high-growth decade of 2000–2001 to 2010–2011 to analyse whether there has been a marked tendency of this ‘traditional’ segment to transform into a ‘modern’ segment. We construct a variable, the net accumulation fund, which indicates the ability of an enterprise to accumulate and grow, and explore its evolution, over time and across industries, for enterprises with different production structures and firm-level characteristics. We show that while, on one hand, the average ‘traditional’ enterprise has been able to economically reproduce itself rather than withering away, the dualism between the ‘traditional’/non-capitalist and the ‘modern’/capitalist segments has been reproduced and further reinforced during this period of high economic growth, raising questions about the process of economic transformation as envisaged in much of development literature.

Keywords Informal sector · Dualism · Structural transformation · Manufacturing · India

Résumé

Nous identifions un dualisme fondamental dans le secteur manufacturier informel (SMI) en Inde entre un segment “traditionnel”/non capitaliste, comprenant des entreprises familiales, basées sur le foyer, qui constituent la grande majorité du SMI, et un segment “moderne”/d’entreprises capitalistes employant de la main d’œuvre salariée. Nous nous concentrons sur la décennie à forte croissance de 2000-01 à

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2010-11 afin d'analyser s'il existe une tendance marquée de ce segment «traditionnel» à se transformer en un segment «moderne». Nous construisons une variable, le fonds d'accumulation net, qui indique la capacité d'une entreprise à accumuler et à se développer, puis nous étudions son évolution, dans le temps et selon les industries, pour des entreprises ayant des structures de production et des caractéristiques au niveau de l'entreprise différentes. Nous montrons que, si, d'une part, l'entreprise "traditionnelle" moyenne a été capable de se reproduire économiquement plutôt que de déperir, le dualisme entre les segments "traditionnel"/non capitaliste et "moderne"/capitaliste a été reproduit et renforcé encore pendant la période de forte croissance économique, soulevant des questions sur le processus de transformation économique tel qu'envisagé dans de nombreux ouvrages sur le développement.

JEL Classification O14 · O17 · J46

Introduction

The informal sector in India has continued to persist in spite of sustained economic growth over the past couple of decades. Following much of the literature on economic modernisation (Lewis 1954; Nurkse 1952; Ranis and Stewart 1999; Rosenstein-Rodan 1943), one would have expected that, with economic growth, the dualism between 'traditional'/non-capitalist and 'modern'/capitalist segments of the economy would wither away. This process may eventually bring about a full-fledged transformation of the entire economic structure, with formalisation of the informal sector over time. However, in spite of some success in industrialisation over the decades and sustained economic growth since the mid-1980s, approximately 93% of total employment and 86% of non-agricultural employment in India still remain informal in nature (Mehrotra et al. 2012). Even within the manufacturing sector—which plays a vitally important role in the transformation process given its strong backward and forward linkages (Storm 2015)—the unorganised/informal segment contributes more than 69% of total employment.

Existing literature does not rigorously analyse the evolution of the informal sector in India to examine whether there has been a tendency towards its transformation during the period of high economic growth. We take up this task in this article, particularly in the context of the informal manufacturing sector (IMS). We show that the IMS is not homogeneous; rather, there exists a basic dualism within the IMS between 'traditional'/non-capitalist and 'modern'/capitalist segments. The continued persistence of this dualism raises questions about the possibility of development via the theoretically envisaged path of transformation. We construct a variable, the net accumulation fund (NAF), which is the fund retained by enterprises for possible accumulation and reinvestment to grow and transition into larger enterprises. We empirically investigate, over time and across industries, the evolution of the NAF of informal manufacturing firms with different types of production structures and labour processes and micro-level enterprise characteristics to explore the possibility



of a dilution of the dualism in the IMS and its implications for the process of structural transformation in India.

The rest of the article is structured as follows: The next section provides a brief review of literature and contextualises the issues to be explored. The following section gives a description of the data sources and definitions used in this study. The next two sections discuss the construction of the NAF and provide an exploratory analysis of the NAF and certain characteristics of the IMS enterprises. In the section on “[Model specification and results](#)”, we carry out a detailed analysis using independently pooled ordinary least-squares (OLS) regressions, Oaxaca–Blinder decomposition, and fixed effects pseudo-panel estimation. The concluding section provides a discussion of the implications of this analysis for understanding the possibilities of transformation of the informal economy in India.

Informal Economy, Dualism and Structural Transformation

Lewis (1954) theorised the process of economic development as a transformation of the economic structure of a developing economy. The developing economy is characterised as distinct from a developed one in terms of its structure. While the former is conceptualised as a dual economy having a small ‘modern’ and a large ‘traditional’ sector marked by the presence of a surplus labour force, the latter is seen to be homogeneously ‘modern’.¹ A structural transformation in the developing economy then entails a transition of the ‘traditional/agricultural/pre-capitalist’ sector into a ‘modern/industrial/capitalist’ sector (Tingor 2004), leading to a dissolution of the economic dualism and a diminution in the size of the surplus labour force (Ghose 2010). India, with its large agricultural and informal sectors and a relatively small formal sector, is a classic example of such a dual economy.

Following Lewis, the issue of structural transformation has spawned a rich literature that interrogates the possibility of this transformation in different economies across the world, often from macro economy-wide perspectives (Binswanger-Mkhize 2013; Timmer and Akkus 2008). Some of this literature, explicitly or implicitly, has dealt with the idea of transformation in terms of the withering away of the informal economy, either by its crowding out or by its transition into the formal sector (see, e.g. Moser 1994). Based on these arguments, the evolution of the informal sector, its interaction with the growth process and the possibilities of its transformation have been widely debated. Some view the informal economy as a residual ‘traditional’ sector in the economy, which is driven out, albeit slowly, as the economy grows and develops, and more people find employment in the more productive formal sector (Bardhan 2009; ILO 2018; Moser 1994; Margolis 2014; Papola 1980). La Porta and Shleifer (2014), through a cross-country study, show that while there

¹ In the Lewisian literature, the ‘traditional’ segment of the economy is seen to be the one that absorbs the surplus labour force in the economy, is relatively stagnant and is less productive, whereas the ‘modern’ segment is viewed as the dynamic, profit-maximising segment that is relatively more productive and technologically advanced.



exists the possibility of such a transformation, this process might be exceedingly slow and the dualism is likely to persist for a long period of time. Others, in contrast, view the informal sector as mainly micro-entrepreneurial, risk-taking, profit-maximising and dynamic in nature, which can itself act as an engine for growth and where individuals voluntarily choose to participate and find employment (De Soto 1989; Fajnzylber et al. 2009; Falco and Haywood 2016; Maloney 2004).

However, Ranis and Stewart (1999), in their seminal article, argue that the informal sector can be quite heterogeneous, consisting of both 'traditional' and 'modern' segments. In other words, there might be a dualism *within* the informal sector itself. They characterise the enterprises in the 'modern' segment as dynamic entities that are relatively technologically advanced, more productive, capital intensive and employing higher-skilled workers, as compared with the relatively stagnant 'traditional' enterprises. The 'modern' segment may have a complementary relation with the formal sector, and growth in the latter, along with its strong and favourable linkages with the former, may lead to crowding-out of the 'traditional' segment of the informal sector by the 'modern' segment. The 'modern' segment may, in turn, be eventually crowded out by the formal sector, thereby bringing about a transformation of the economic structure. Several recent empirical studies across developing economies have found evidence of heterogeneity within the informal sector that may consist of both micro-entrepreneurial as well as residual segments (García 2017; Günther and Launov 2012; Nguyen et al. 2013; Radchenko 2014; Mandelman and Montes-Rojas 2009).

A recent strand of literature characterises the 'modern' segment of the informal sector as comprising capitalist enterprises, which, it argues, are governed by the economic logic of generating surplus for accumulation and growth, with clear demarcations between the household space of consumption and the enterprise space of production, while the 'traditional' segment comprises non-capitalist enterprises, which are mainly governed by the subsistence needs of the households running them, with overlap between the consumption and the production spaces of the household and the enterprise (Sanyal 2007; Bhattacharya et al. 2013; Bhattacharya 2017; Bhattacharya and Kesar 2018). In contrast to the expectation of transformation within the informal economy—as well in the overall economic structure—as posited in some of the literature discussed above, this strand is pessimistic about the possibilities of such transformation in developing economies, given the serious lack of alternative livelihood opportunities for households engaged in the 'traditional' segment of the informal economy and the exclusionary nature of growth processes in such economies (Bhaduri 2017; Harriss-White 2012; Sanyal 2007).

The informal sector in India has remained an important source of livelihood for the vast majority of the workforce in spite of a long-drawn process of industrialisation and an acceleration of growth since the mid-1980s (Chakrabarti 2016; Raj and Sen 2016). During the past decade (2000–2001 to 2010–2011), which is widely identified as a period of high growth in the Indian economy, there has actually been an absolute increase in the total number of people employed in the unorganised/



informal sector (from about 343 to 387 million), while its distributional share in the total employment has declined only marginally from about 86% to 84% (Mehrotra et al. 2012).² Even within the manufacturing sector, as noted above, more than two-thirds of the workforce has been employed in the unorganised/informal segment.

We argue in the next section that the Indian IMS is not a homogeneous sector—rather it comprises a large ‘traditional’/non-capitalist segment and a much smaller ‘modern’/capitalist segment (we define these terms, outlining their specificities as we understand them for the purpose of this paper, in the next section). A detailed and rigorous analysis of the continued persistence of this dualist informal economy in India and the possibilities of a transformation of its structure during the period of high economic growth is largely missing from literature.³ While the issue of transformation has been studied in varied ways in literature as outlined above, for the purpose of this paper, we focus on a particular aspect of transformation, specifically in the context of the informal economy. By transformation of the dualist structure of IMS in India, we refer to a dilution of the ‘modern’–‘traditional’ dichotomy within the sector. Such a dilution might occur if either the ‘traditional’ segment is not able to economically reproduce itself and withers away over time, or the ‘traditional’ enterprises, on average, are able to grow and expand into dynamic, ‘modern’ enterprises, leading to a dissipation of the dichotomy in the structure of the IMS.⁴ While this transformation may not come to pass within a decade, in this analysis, we explore whether there has been a marked *tendency* towards such transformation in the Indian IMS over the past decade, particularly since this process was expected to be hastened with high economic growth.

² The resulting marginal rise in employment in the organised sector as a proportion of total employment has, however, been mainly attributed to the increase in the ‘informally employed’ workforce within the organised sector (Ghose 2015; Srivastava 2012).

³ While Raj and Sen (2016) present an analysis of the evolution of the Indian IMS and Nataraj (2011) analyses the impact of trade liberalisation on the Indian IMS, they do not rigorously explore the transformative possibilities of this sector.

⁴ It may be argued that traditional household enterprises voluntarily *choose* to remain small in order to avoid various kinds of labour regulations, which they would have to abide by if their size exceeded a specific threshold level (Perry 2007). For example, in the Indian context, if a firm employs more workers than a particular threshold level, it will be considered to be part of the formal sector of the economy and, under specific circumstances, will be legally bound to enforce various kinds of labour regulations and environmental standards. Implementing such measures might increase the cost of business for the firm. Firms might voluntarily choose to remain under such specific threshold levels in order to avoid these regulations. However, the issue we are exploring here is distinctly different from such cases. Here, we look at the possibilities of transformation *within* the IMS, i.e. a transition of the ‘traditional’ segment of the IMS into ‘modern’, dynamic enterprises, rather than a transition from the informal to formal sector. The question of voluntarily choosing to remain small in order to avoid costly regulations might be an issue for the larger enterprises in the IMS, e.g. for those at the upper end of the ‘modern’ segment of the IMS. The enterprises in the ‘traditional’ segment of the IMS, on the other hand, do not employ *any* wage labour and the sizes of such enterprises is nowhere near the threshold levels. Such ‘traditional’ enterprises constitute the overwhelming majority (about 85%) of the IMS.



Definition and Data Sources

The National Commission for Enterprises in the Unorganised Sector (NCEUS) provides an internationally comparable definition of the informal sector in the Indian context: “The unorganised (informal) sector consists of all unincorporated private enterprises owned by individuals or households engaged in the sale and production of goods and services operated on a proprietary or partnership basis and with less than ten total workers” (NCEUS 2007, p. 48). However, the National Sample Survey Organisation (NSSO), whose surveys provide firm-level data on the IMS, uses the category of ‘unorganised sector’, which includes even larger enterprises, as well as other non-household (neither proprietary nor partnership) enterprises.⁵ For the purpose of the analysis carried out here, we rely on the NCEUS definition of the informal sector and extract the data from NSSO surveys of unorganised manufacturing enterprises, by limiting the set of enterprises to those which conform to the official NCEUS definition.⁶

The NSSO collects data on two types of unorganised manufacturing enterprises which constitute this sector, marked by distinct production structures and labour processes: own account manufacturing enterprises (OAMEs) run without any regular hired workers, and establishments that employ at least one regular hired worker.⁷ The OAMEs, accounting for 85% of IMS enterprises, are mostly petty production units that are technologically less advanced, employ fewer assets, have low levels of capital and are mostly located within households. We characterise these enterprises as the ‘traditional’ segment of the IMS. While the establishments are also small enterprises, they are relatively much more productive and dynamic, employ more assets and higher levels of capital compared to the OAMEs, and are mostly located outside households, and can be understood as the ‘modern’ segment. Following the recent strand of literature discussed in the previous section (Sanyal 2007; Bhattacharya 2017), in the context of the Indian IMS, the OAMEs can also be characterised as non-capitalist enterprises, which are marked by an absence of capital–wage labour relation and are mainly involved in petty commodity production mostly with the help of unpaid family labour. The establishments, on the other hand, can be characterised as capitalist enterprises, marked by the existence of wage labour and, as mentioned above, characteristically different from the petty production units. It should be noted

⁵ The official definition recognises the unorganised segment as constituting the following enterprises: “(i) All manufacturing enterprises except those registered under section 2m(i) and 2m(ii) of Factories Act, 1948 and Bidi and Cigar Workers (conditions of employment) Act, 1966. (ii) All manufacturing enterprises except those run by Government (Central Government, State Governments, Local Bodies)/ Public Sector Enterprises” (NSSO report 525, 2008, p. 8).

⁶ We find that, for all three time points of our analysis (2000–2001, 2005–2006 and 2010–2011), the informal enterprises account for around 95–98% of the estimated population of unorganised enterprises—and around 92–96% of the sample—surveyed by the NSSO.

⁷ The NSSO further categorises the establishments into two types: (a) non-directory manufacturing enterprises (NDMEs) employing one to five workers, along with at least one hired worker, and (b) directory manufacturing enterprises (DMEs) employing six to nine workers with at least one hired worker. We have carried out the analysis in this paper with this further categorisation as well, but found that it does not change the results or add any new insights.



that, while production/labour processes marked by the existence/absence of hired wage labour is a defining distinction between OAMEs and establishments, our paper shows that the characteristics of these two categories of enterprises also correspond to the distinctions outlined in other strands of literature referred to above.

This analysis uses unit-level data from the three rounds of the NSSO surveys of the unorganised manufacturing sector, covering the high growth decade of 2000–2001 to 2010–2011 (56th round for 2000–2001, 62nd round for 2005–2006 and 67th round for 2010–2011). For this analysis, all monetary values are indexed to real terms at 2004–2005 prices, using the gross domestic product (GDP) deflator of the unorganised manufacturing sector.

Net Accumulation Fund

A unique characteristic of the IMS is an overlap between enterprises and households, since either they engage only family labour—specifically the OAMEs, which constitute roughly 85% of the IMS—or they have both hired labour and family labour (including working owners). There is no explicit payment of wages to the family labour. The net retained earning (NRE) of the enterprise, which is the gross value added (GVA) minus the wages of hired workers and rents and interest payments, is also the net income of the household from the enterprise. IMS enterprises usually do not have access to formal credit, and informal credit is often too expensive (which might be the reason why NSSO data shows that most IMS enterprises do not have significant loans). Therefore, the growth of such enterprises generally depends on their ability to retain sufficient funds for accumulation from the NRE, after keeping aside an amount for the unpaid family labour and working owners of the enterprise. This amount can be seen as a ‘pseudo-wage’ fund that the enterprise would have needed to pay to the unpaid household workers and working owners had they been hired as wage workers in a such an enterprise. This fund might also be interpreted as the amount that the household enterprise might possibly keep aside in order to meet the consumption needs of the unpaid family labour and working owners. In order to get a sense of the fund left in the enterprise for possible accumulation and reinvestment, we deduct the ‘pseudo-wage’ fund from the NRE of the enterprise. While the NSSO surveys provide data on the household enterprise’s receipts, expenses and wages for hired workers, there is no reported data on the amount that the enterprise might set aside for the unpaid family labour and working owners. Hence, to estimate the fund that might possibly be retained by these enterprises for its accumulation and growth, we need to estimate the ‘pseudo-wage’ fund and deduct it from the NRE of the enterprises.

For this purpose, we impute ‘pseudo-wages’ for the non-hired workers and the working owners of the firm. For establishments, which hire at least one worker but do not report a wage for non-hired workers, the average wage per hired worker is assigned as a ‘pseudo-wage’ for each non-hired worker and working owner. This ‘pseudo-wage’ is multiplied by the total number of non-hired workers and working owners of the establishment to obtain an estimate of the ‘pseudo-wage’ fund for them.



However, since OAMEs do not employ any hired workers, we assign the average wage per hired worker from a ‘similar’ establishment as the ‘pseudo-wage’ for the non-hired workers and working owners of the OAMEs. We find a ‘similar’ establishment at the same time point on the basis of four parameters: firm size (considering only establishments that employ only one hired wage worker), GVA per worker of the enterprise, sector (rural or urban) and industry type [according to National Industrial Classification (NIC) codes]. We divide the entire data for each time point over the decade (2000–2001, 2005–2006 and 2010–2011) into groups of firms (or cohorts) with same two-digit NIC code and belonging to the same sector, forming 48 cohorts (24 NIC codes $\times$ 2 sectors) for each time point. In order to find a ‘similar’ establishment for an OAME, we compare the GVA per worker of the OAME with establishments that employ only one hired worker and belong to the same cohort. If the GVA per worker of the OAME matches that of such an establishment, we assign the average wage per hired worker in that establishment as the ‘pseudo-wage’ of the working owners and unpaid family workers in the OAME. If it does not match, the ‘pseudo-wage’ for a worker engaged in the OAME is assigned by interpolating or extrapolating the average wage per worker paid in the nearest establishment with one hired worker in the same cohort. This interpolation/extrapolation is done based on the ratio of wages to GVA per worker in the establishment that is nearest to the OAME in terms of GVA per worker. This ‘pseudo-wage’ per worker is multiplied by the total number of non-hired workers (including the working owner) to obtain an estimate of the ‘pseudo-wage’ fund for the working owners and the unpaid family workers in the OAME. The average (median) estimated wages/‘pseudo-wages’ of workers in establishments and OAMEs are reported in Table 1.

Once we have estimated the ‘pseudo-wage’ funds for the enterprises (OAMEs as well as establishments), this fund is subtracted from the NRE of the household enterprise to construct a new variable—the *net accumulation fund* (NAF), which is the fund that the enterprise may use for possible accumulation (or reinvestment) and growth.⁸

Therefore, the NAF

$$\begin{aligned}
 &= \text{GVA} - \text{wage fund of hired workers} - \text{rent} - \text{interest} - \text{‘pseudo-wage’ fund} \\
 &\text{of non-hired family labour and working owners} \\
 &= \text{NRE} - \text{‘pseudo-wage’ fund (alternatively termed the consumption fund)} \\
 &\text{of non-hired family labour and working owners}
 \end{aligned}$$

⁸ The NAF is distinct from the usual notion of NRE or net profit for enterprises where the entire production is carried out using hired labour. For such enterprises, the net profit (which is also the fund for accumulation and reinvestment) is calculated by deducting from their GVA the wages paid to hired workers, as well as rents and interest payments. However, as noted above, the vast majority of informal enterprises carry out production without any hired labour, and even when some wage labour is hired, the household members still participate in the production process without receiving any wages. Hence, to calculate the NAF for such enterprises, we estimate the ‘pseudo-wage’ fund for the unpaid household workers, and then deduct it from the NRE.



Table 1 Median wages/'pseudo-wages' of workers in establishments and OAMEs (real annual values in INR)

	Rural			Urban		
	2000–2001	2005–2006	2010–2011	2000–2001	2005–2006	2010–2011
Establishments	12,277	15,084	22,638	20,461	20,993	30,561
OAMEs	5184	5319	7984	7735	7212	11,976

It is possible that the working owners and other unpaid family workers might retain a higher fund than is paid to the wage workers. Further, in addition to the wages imputed to them for their labour, there might also be a return to managing the firm. However, by using the wage of the hired worker as a yardstick to estimate the 'pseudo-wage' fund, we are able to estimate the part of the NRE that the enterprise *might* be able to retain as NAF, even if the 'pseudo-wages' assigned to the working owners and unpaid family labour are as low (or as much) as the wages of the hired workers in that (or a 'similar') enterprise. It should be noted that, since the household has the right over the NRE, it may decide to consume or reinvest this earning in any proportion.⁹ Similarly, it is possible that the working owner may receive a higher fund than what is paid to the wage worker—a difference that may be attributed to the return to managing the firm. However, as noted above, the *estimated* NAF indicates the possibilities for accumulation and growth of the enterprise even if the funds kept aside for the working owners and unpaid family labour were similar to those of the wage workers.

Evolution of Net Accumulation Fund

A cursory glance at the median annual real NAF for the two types of enterprises (OAMEs and establishments) at the three time points (2000–2001, 2005–2006 and 2010–2011) shows that, even though the NAF rose over the considered decade for all types of enterprises, the rise for the OAMEs was not smooth, with a fall in the first half of the period (Table 2). The decadal growth rate was also substantially lower for the OAMEs (about 57%) compared with establishments. Moreover, in 2010–2011, even at the upper end of the NAF distribution for the OAMEs, the monthly NAF retained was very low—INR 1148 (USD 102) at the 75th quantile and INR 2242 (USD 200) at the 90th quantile for OAMEs—whereas the corresponding values for establishments were INR 7621 (USD 679) and INR 14,136 (USD 1259),

⁹ Note that the wage fund of the hired workers may also have a savings component, and hence assigning the entire 'pseudo-wage' fund as the consumption fund for the household workers may be an overestimation of the consumption fund. However, given the extremely low levels of wages of informal wage workers, the saving component is likely to be negligible.



Table 2 Evolution of median net accumulation fund (NAF) (real annual values in INR)

Type of enterprise	2000–2001	2005–2006	2010–2011	Growth rate over decade (%)
OAMEs	3943	2978	6183	56.81
Establishments	23,770	25,892	44,461	87
Total	4638	3650	7547	62.72

respectively (Table 3).¹⁰ This low capacity to retain NAF by OAMEs indicates their weak position in terms of expanding and growing into establishments.¹¹ However, we also find that, on an average, they retain a positive, though trivial, amount of NAF. This implies that, instead of withering away, the OAMEs have been able to economically reproduce themselves over time, even though at very low levels.

Moreover, the *absolute* difference in median NAF between OAMEs and establishments rose over the decade (Fig. 1). This difference grew by about 16% in the first half of the decade, which accelerated to 67% over the later half, with the difference being more pronounced in urban areas. This is indicative of a starkly widening gap between the ‘modern’ and ‘traditional’ segments. Also, the difference in values between the upper and lower quantiles of the NAF distribution markedly increased over the decade for each type of enterprise (OAMEs or establishments).

To further explore the evolution of the NAF, we consider the following set of enterprise characteristics: (a) sector (rural or urban), (b) location (inside or outside the household), (c) registration (registered or unregistered), (d) accounts maintenance (maintained or not), (e) total value of assets held, (f) subcontracting linkages (whether subcontracted or not) and (g) gender of owner (female- or non-female headed). These characteristics can be understood to represent various ‘favourable’ or ‘unfavourable’ socio-economic dimensions of enterprises that may affect their performance and functioning. First, urban location is considered to be a relatively favourable characteristic for an enterprise, given the possibility of better access to markets. Second, enterprises located outside the household, those which are registered and those maintaining accounts are considered to be better positioned compared with their counterparts. Moreno-Monroy et al. (2014) calculate the degree

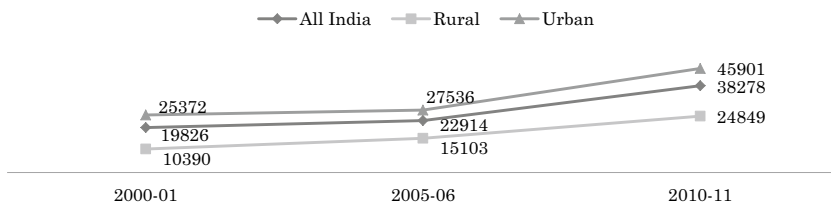
¹⁰ Given the high level of heterogeneity among the informal firms and, consequently, the high dispersion in the data, the averages are reported at median values throughout this section, unless specified otherwise. For reference purposes, all monetary values can be converted from Indian rupees (INR) to US dollars (USD) following the purchasing power parity (PPP) conversion rate (INR/USD) of 11.23 for the years 2004 and 2005 on an average (11.171 for 2004 and 11.282 for 2005) (OECD 2017).

¹¹ Note that the establishments in the lowest decile of the NAF distribution, on an average, have negative NAFs. This is due to the fact that, on an average, the wages of hired workers for these establishments as well as the average amount of non-hired family labour working in these enterprises are anomalously higher than those for the entire set of establishments, thereby raising the values of imputed ‘pseudo-wage’ funds and leading to negative NAFs. However, the average GVA for these establishments is much higher than that of OAMEs with negative NAFs for corresponding time points. Therefore, even with negative NAFs, such establishments have a much larger scale of operation.



Table 3 NAF of OAMEs and establishments over quantiles (real annual values in INR)

NAF	OAMEs			Establishments		
	2000–2001	2005–2006	2010–2011	2000–2001	2005–2006	2010–2011
p10	517	448	1233	–5456	–6415	3350
p25	1460	1024	2493	7305	7348	18,292
p50	3943	2978	6183	23,769	25,892	44,461
p75	9455	8112	13,779	54,278	58,046	91,458
p90	17,802	16,561	26,909	102,035	116,629	169,635

**Fig. 1** Difference in median NAF of establishments and OAMEs (real annual values in INR)

of ‘modernity’ of an enterprise based on its location, where a fixed location outside the household is considered to be a relatively favourable characteristic. They find that such enterprises, on an average, employ more workers, have higher value-added and possess more assets as compared with those located within the household (‘traditional’ ones). Also, enterprises located outside the household are expected to have better access to markets (Basole et al. 2015). Further, registered enterprises are expected to be more stable, and those that maintain accounts are expected to employ more human capital (Raj and Sen 2016). Finally, enterprises with higher assets value, those which are non-female headed and those that are non-subcontracted are considered to be socio-economically stronger. Availability of assets is an important indicator of the sustainability and stability of enterprises (Basole et al. 2015); female-headed enterprises are expected to be socio-economically weaker than non-female-headed ones given the broader patriarchal socio-cultural structure (Sethuraman 1998); and it has been empirically shown in literature that subcontracted enterprises in the Indian IMS, on an average, are much less productive than non-subcontracted ones (Basole et al. 2015; Bhattacharya et al. 2013; Raj and Sen 2016). At a preliminary level, we find that a much higher proportion of establishments exhibit these ‘better/favourable’ characteristics relative to OAMEs, with this difference in the proportions either rising or continuing to persist over the decade (Tables 4, 5).

We further analyse these issues in the following section, which provides a detailed econometric analysis of the evolution of the NAF between OAMEs and establishments.



Table 4 Percentage of OAMEs and establishments displaying ‘favourable’ characteristics

	Located outside the HH		Maintained accounts		Registered	
	OAMEs	Establishments	OAMEs	Establishments	OAMEs	Establishments
2000–2001	22.92	73.93	0.60	6.98	4.94	39.27
2005–2006	19.30	75.01	3.21	15.65	4.72	38.29
2010–2011	20.13	74.45	1.71	20.25	7.48	47.59

Table 5 Percentage of OAMEs and establishments showing ‘unfavourable’ characteristics

	Female-headed		Subcontracted	
	OAMEs	Establishments	OAMEs	Establishments
2000–2001	29.60	4.84	30.70	30.37
2005–2006	41.27	5.94	32.00	27.33
2010–2011	45.42	5.23	22.25	9.88

Model Specifications and Results

This analysis is divided into three parts that empirically investigate the evolution of the NAF over our period of analysis. The first part employs an OLS regression framework of independently pooled cross-sections of enterprises over three rounds of NSSO data to explore the evolution of the NAF for OAMEs vis-à-vis establishments, along with different enterprise characteristics. These regressions are also done separately for each of the eight major industries in the IMS as per the National Industrial Classification (NIC), and also separately for high growth and low growth states.¹² In the second part, we carry out an Oaxaca–Blinder decomposition to analyse what part of the observed difference in the NAF between OAMEs and establishments can be explained by the difference in their enterprise characteristics, and what part of the difference may be attributed to the ‘differential effects’ of these characteristics on these two types of enterprises (or to some other unobserved factors). Finally, in the third part, we construct a cohort-level pseudo-panel to analyse the difference in the evolution of the NAF between OAMEs and establishments for

¹² The eight major NIC codes (NIC 2004 classification) in the Indian IMS are manufacture of wearing apparel, dressing and dyeing of fur (NIC 18); manufacture of food products and beverages (NIC 15); manufacture of tobacco products (NIC 16); manufacture of textiles (NIC 17); manufacture of wood and products of wood and cork, except furniture; manufacture of articles of straw and plating materials (NIC 20); manufacture of fabricated metal products, except machinery and equipment (NIC 28); manufacture of furniture, manufacturing not elsewhere classified (NIC 36); manufacture of other non-metallic mineral products (NIC 26). These eight industries together account for more than 90% of the IMS (varying from 92% to 94% over the three rounds).



the same cohort over time, as well as to control for any unobserved time-invariant heterogeneity. This further acts as an additional robustness check for our analysis.

Pooled OLS Regression

To explore the evolution of the NAF for OAMEs and establishments along with the enterprise characteristics, the following two equations are estimated:

$$NAF_i = \alpha + \beta \text{Enterprise-type}_i + \gamma \mathbf{X}_i + \Gamma \mathbf{Z}_i + \lambda \text{Year}_i + \varepsilon_i, \quad (1)$$

$$NAF_i = \alpha' + \beta' \text{Enterprise-type}_i + \gamma' \mathbf{X}_i + \Gamma' \mathbf{Z}_i + \lambda' \text{Year}_i + \Psi' \text{Enterprise-type}_i * \text{Year}_i + \varepsilon'_i, \quad (2)$$

where, for firm i , NAF is a proxy for the firm's ability to accumulate and grow.¹³ 'Enterprise-type' is a dummy variable that takes a value of 1 for an establishment and 0 for an OAME. $\mathbf{X}$ is a vector of enterprise characteristics (discussed above), and $\mathbf{Z}$ is a vector of control variables, which include (a) enterprise-level controls: (a-i) activity type, i.e. whether one or more than one activity (referred to as mixed activity) is carried out by a single enterprise; (a-ii) nature of operation, which captures the seasonality of the enterprise, i.e. whether the enterprise operated more or less regularly throughout the year (perennial), or whether it operated only for particular seasons in a year (seasonal), or whether it operated occasionally, but for more than 30 days in a year (casual); and (b) state- and industry-level controls: (b-i) state dummies to control for state fixed effects; (b-ii) NIC codes to control for industry fixed effects; (b-iii) per capita gross state domestic product; (b-iv) percentage of manufacturing domestic product in total state domestic product; (b-v) state-level net sown area as a percentage of total geographical area in the state; (b-vi) state-level workforce participation rate in manufacturing sector (per 1000); (b-vii) state-level infant mortality rate (per 100); (b-viii) state-level literacy rates (per 100).¹⁴ 'Year' is a time dummy for the year of the survey, with 2000–2001 as the base category. These equations are estimated using an OLS regression framework on independently pooled cross-sections of enterprises at three time points (2000–2001, 2005–2006 and 2010–2011) over the decade, using time (Eq. 1) and time interaction (Eq. 2) dummies.

¹³ One can possibly take a log transformation of the NAF, given the skewed nature of the NAF distribution. However, we refrain from such log transformation in our analysis because: (i) the dependent variable in the regression analysis, the NAF, has many (and quite large) negative values. Out of 384,698 sample observations, 21,190 observations (i.e. approximately 5% of the sample) have a negative NAF. Taking a log would automatically truncate our distribution, but we would ideally not want to lose information on these observations, as they can have important implications for the results; and (ii) while log transformation may bring normality to the distribution, given the large sample size, the 'large sample properties' would be applicable for the data, and thus we do not need to assume normality of distribution for any estimations.

¹⁴ Per capita state domestic product for the three time points would control for the overall economic performance of the states. State-level literacy rate and infant mortality rate would control for social infrastructure in the state. The percentage of land under agriculture, share of manufacturing domestic product in the total state domestic product and percentage of workforce in the state employed in manufactur-



An estimation of the first equation (Regression 1) allows us to see how the overall difference in NAF between OAMEs and establishments, and between relatively ‘favourable’ and ‘unfavourable’ characteristics, evolved over the decade. However, this regression does not capture how the difference in NAF between OAMEs and establishments changed between the three time points of our analysis. This is analysed in the second regression (Regression 2), which includes an interaction of the enterprise type with time, in addition to the set of explanatory variables included in Regression 1. The results of these regressions (coefficient values and the cluster-robust standard errors for each variable in vector $\mathbf{X}$) are reported in Table 6.

The following interpretation of these results is based on specifications of these regressions that use a complete set of state and NIC controls, i.e. Model Specification 1.2 of Regression 1 and Model Specification 2.2 of Regression 2, as reported in the second and fourth column of Table 6, respectively. The regressions are robust to alternate specifications.

We note from Regression 1 that, on an average, the NAF rose for enterprises over the decade. However, on an average, there was a huge difference in the NAF of an OAME and an establishment, with the NAF in an establishment being approximately INR 23,565 (USD 2098) higher than that in an OAME—this difference itself being around three times the mean NAF retained by OAMEs (refer to Regression 1). Moreover, over the decade, this gap continued to rise steeply (Regression 2). For example, while in 2000–2001, on an average, the NAF of an establishment was approximately INR 15,181 (USD 1352) higher than that of an OAME, *ceteris paribus*, this gap increased to about INR 26,519 (USD 2361) in 2005–2006 and further to INR 28,756 (USD 2561) in 2010–2011—with the difference rising by almost 89% over the decade (refer to Model Specification 2.2, Table 6).¹⁵ This shows a divergence between the establishments (capitalist enterprises) and the OAMEs (non-capitalist enterprises) and, hence, a likely entrenchment of dualism within the informal sector over the decade of high economic growth.

Further, we explore how the NAF varies with the enterprise characteristics outlined above. We find that the NAF of enterprises located in urban areas is, on an average, about INR 2462 (USD 219) higher than those in rural areas, *ceteris paribus* (Model Specification 1.2). Moreover, a much higher proportion of establishments and a relatively lower proportion of OAMEs are located in urban areas.

Footnote 14 (continued)

ing sector would control for extent of and dependence on agriculture and manufacturing activities in the state. Skill level of workers might be another important variable to control for. However, as pointed out by Raj and Sen (2016), urban versus rural location might be a good proxy for skill. Additionally, skills are expected to vary widely across industry groups, which we control for by adding NIC-level controls in the regression. We are unable to control for state-wise regulatory environments due to unavailability of data for all states for our period of analysis. However, any state- or industry-level policy/regulation would already be controlled for by the state and NIC dummies that account for any state- or NIC-level fixed effects.

¹⁵ We obtain the figures for the differences between the NAF of OAMEs and establishments for 2005–2006 and 2010–2011—INR 26,519 and INR 28,756, respectively—by adding the coefficient for the enterprise type dummy for the base time period (i.e. 15,181) and that for the interaction between the enterprise type dummy and the time dummy (i.e. 11,338 for 2005–2006 and 13,575 for 2010–2011).



Table 6 Pooled OLS regression (all India)—(dependent variable: NAF)

	Regression 1		Regression 2	
	Model Specification 1.1	Model Specification 1.2	Model Specification 2.1	Model Specification 2.2
Establishment (base category: OAME)	24,232.5 ^{***} (2894.1)	23,565.2 ^{***} (2766.0)	15,961.5 ^{***} (2599.1)	15,181.2 ^{***} (2202.6)
Establishment#Time 2005–2006			11,319.7 [*] (5721.9)	11,337.7 ^{**} (5405.5)
Establishment#Time 2010–2011			13,409.7 ^{***} (1660.7)	13,574.8 ^{***} (1758.6)
Time dummy 2005–2006 (base category: year 2000–2001)	203.7 (769.7)	188.3 (1299.0)	–1241.9 ^{***} (410.1)	–1669.1 [*] (938.7)
Time dummy 2011–2012 (base category: year 2000–2001)	3596.4 ^{***} (679.4)	4597.4 [*] (2620.0)	1797.7 ^{***} (539.4)	1161.8 (2454.0)
Urban (base category: rural)	3026.9 ^{***} (746.6)	2462.0 [*] (1365.0)	3188.3 ^{***} (764.1)	2521.1 [*] (1293.4)
Value of assets held (in INR)	0.0353 ^{***} (0.00431)	0.0342 ^{***} (0.00438)	0.0349 ^{***} (0.00425)	0.0338 ^{***} (0.00432)
Contract dummy (base category: non-subcontracted)	–2407.1 ^{***} (696.9)	–2235.5 [*] (1263.1)	–2311.7 ^{***} (706.8)	–2147.5 (1277.9)
Non-female-headed enterprise (base category: female-headed enterprise)	5045.6 ^{***} (536.6)	4460.9 ^{***} (374.0)	4881.7 ^{***} (537.0)	4272.3 ^{***} (362.3)
Located outside household (base category: located within HH)	2841.4 ^{***} (562.9)	1967.5 ^{***} (422.0)	2872.0 ^{***} (564.6)	1977.3 ^{***} (424.5)
Enterprise registration (base category: unregistered)	13,031.3 ^{***} (1599.5)	12,372.4 ^{***} (1543.0)	13,063.8 ^{***} (1680.9)	12,437.3 ^{***} (1609.0)
Accounts maintained by enterprise (base category: accounts not maintained)	43,865.8 ^{***} (7576.0)	42,919.3 ^{***} (7537.0)	42,996.4 ^{***} (7557.0)	42,022.6 ^{***} (7521.6)



Table 6 (continued)

	Regression 1		Regression 2	
	Model Specification 1.1	Model Specification 1.2	Model Specification 2.1	Model Specification 2.2
Mixed activity (base category: enterprise does not undertake mixed activity)	3091.4** (1435.4)	2922.1** (1327.7)	3226.9** (1467.8)	3027.1** (1347.8)
Nature of operation: perennial (base category: seasonal/casual)	6303.9*** (642.2)	6895.1*** (833.5)	6351.6*** (668.5)	6920.5*** (826.7)
Per capita gross state domestic product (GSDP) (in INR)		0.0765 (0.0466)		0.0609 (0.0388)
Proportion of manufacturing GDP in total state GDP		131.2 (358.2)		175.6 (369.9)
Agriculture level in state: (net sown area as percentage of total geographical area in state)		-35.28 (152.8)		-6.373 (144.1)
State-level workforce participation ratio in manufacturing sector (per 1000)		-0.745 (9.524)		-0.439 (9.165)
State-level infant mortality rate (per 100)		22.80 (92.60)		-5.370 (94.74)
State-level literacy rate (per 100)		-197.0 (174.4)		-71.75 (131.8)
State controls	No	Yes	No	Yes
NIC controls	No	Yes	No	Yes
Constant	-5546.2*** (793.5)	10,563.8 (14,755.9)	-4468.5*** (789.8)	5266.4 (12,525.4)
Observations	382,993	382,993	382,993	382,993

Robust standard errors in parentheses clustered at state level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ 

We also find that, on an average, all else being equal, an increase in a firm's asset value by INR 10,000 increases its NAF by about INR 342 (Model Specification 1.2). However, while the median value of assets of OAMEs increased from INR 11,594 to INR 30,184 (from approximately USD 1032 to USD 2688) over the decade, the asset value of establishments increased from INR 129,817 to INR 258,602 (from about USD 11,560 to USD 23,028), resulting in a widening gap in the NAF of OAMEs and establishments over time. Moreover, the reported value of assets of the OAMEs is usually inflated, since most OAMEs operate from within households, and the value of the household space cannot be considered strictly to be a part of the asset value of the enterprise, since it is also used as the owners' residence. Therefore, the increase in NAF is likely to be even higher for establishments than for OAMEs over the decade.

Another important characteristic of an enterprise is its subcontracting linkage. We find that, on an average, the NAF of a subcontracted firm is INR 2236 (USD 199) less than that of a non-subcontracted one (Model Specification 1.2)—this difference is about 23% of the mean NAF of the subcontracted and 12% of the non-subcontracted enterprises over the decade. While the incidence of subcontracting fell over the decade for both OAMEs (from 30% to 22%) and establishments (from 30% to 10%), the fall was much more pronounced for establishments. Therefore, the relatively worse outcome associated with being subcontracted remained more a feature of OAMEs than establishments.

Further, it is found that the NAF of a female-headed firm, on an average, is about INR 4461 (USD 397) less than that of a non-female-headed one, *ceteris paribus* (Model Specification 1.2)—this difference is about 89% of the mean NAF of the female-headed and 20.7% of the non-female-headed enterprises over the decade. However, for OAMEs, the proportion of female-headed enterprises rose steeply over the decade (from 29% to 45%), while it continued to remain at a very low level (approximately 5%) for establishments.

We also find that the NAF of a firm located outside the household, on an average, is approximately INR 1968 (USD 175) more than that of one located within a household, all else being equal (Model Specification 1.2)—this difference is about 6% of the mean NAF of the enterprises located outside the household and 23% of those located within a household over the decade. In a similar vein, the NAF of a registered firm or a firm that maintains accounts (both 'favourable' characteristics), on an average, is much higher than that of a non-registered firm or that of a firm that does not maintain accounts, respectively. However, over the decade, the proportion of OAMEs that exhibited these 'favourable' characteristics remained quite low (e.g. only 19–23% of OAMEs were located outside the household and 5–8% were registered during the decade), whereas the proportion of establishments exhibiting these 'favourable' characteristics remained relatively much higher (with 74–75% of



establishments being located outside the household and 38–48% being registered during the decade) (Table 4).¹⁶

Further, in order to explore the relation with economic growth in detail, we do the above exercise separately for high- and low-growth states. We divide the states into these two sets based on the decadal rates of growth in their per capita state domestic product. We ran a regression of Specification 2.2 of Regression 2, separately for high- and low-growth states, and find that that, while for low-growth states the difference in NAF between establishments and OAMEs rose by almost 60% over the decade, for high-growth states it rose by almost 111%. In other words, the divergence between establishments and OAMEs became more accentuated in states that experienced higher growth.¹⁷

We also carry out Regression 1 (as specified earlier) for the eight major NIC codes separately. We find that the difference in the NAF between establishments and OAMEs remains positive and significant for all industries; i.e. these industry-wise results are similar to the overall results reported above. While for certain NICs the effects of some enterprise characteristics are insignificant, for those that are significant, except for manufacturing of tobacco products (NIC 16), they continue to have the same relation (positive or negative) as the overall results.¹⁸

We find from this analysis that, over this decade of high economic growth, there was an increasing divergence between the NAF of establishments and that of OAMEs. Further, enterprises with ‘favorable’ characteristics, on an average, had much higher NAF than those with the relatively ‘unfavourable’ ones, and a much

¹⁶ While the amount of loans taken by the enterprise would be an important variable that we might need to control for in our regression, we are unable to do so since 93.7% of population in our data do not report any loans. This missing information could either imply that these enterprises do not take loans or have not reported the loans taken. Given this ambiguity, we do not include this variable in our regression (if we were to run the regression without these missing observations, we would only use 6.3% of the population). Additionally, we also ran a regression controlling for land distribution in the state. However, data for Jharkhand are not available for the years 2000–2001 and 2005–2006, thus we do not include this variable as a control. We ran an additional regression, including this control, for all states except Jharkhand, and find that our results hold.

¹⁷ These values are calculated from Specification 2.2 of Regression 2 (i.e. the specification with enterprise type and time interaction dummies and the complete set of controls) that we now run separately for high- and low-growth states. We find that, for low-growth states, the difference in the NAF between establishments and OAMEs in 2000–2001 was INR 15,380 ($p < 0.001$) and increased by INR 9274 between 2000–2001 and 2010–2011 ($p < 0.01$), with the difference in the NAF between the two types of enterprises in the low-growth states increasing by 60% over the decade. For the high-growth states, on the other hand, the difference in the NAF between establishments and OAMEs in 2000–2001 was INR 15,011 ($p < 0.001$) and increased by INR 16,725 between 2000–2001 and 2010–2011 ($p < 0.001$), with the difference in the NAF between the two types of enterprises in the high-growth states increasing by 111% over the decade. We do not report the entire table here due to scarcity of space. As an extension of this work, it would be interesting to see how the structure of dualism evolved in high- versus low-growth periods in India, as well as across high- and low-growth states over these two periods. However, such an analysis lies beyond the scope of this paper.

¹⁸ We find that, for the tobacco industry, unlike the IMS in general, being subcontracted is positively related to the NAF of the enterprise. The tobacco industry also stands out from other industries in terms of the large-scale outsourcing that is prevalent within the industry. Around 70–80% of informal tobacco manufacturing enterprises are subcontracted enterprises, while only 20–31% of the overall IMS enterprises were subcontracted over the decade.



higher proportion of establishments relative to OAMEs exhibited these ‘favourable’ characteristics. Also, the growth in the proportion of characteristically ‘better’ enterprises was much faster for establishments than for OAMEs. This indicates that the divergence in the accumulation possibilities of OAMEs and establishments is further underscored in terms of their enterprise characteristics.

Decomposition of Difference in NAF

Here we use a twofold Oaxaca–Blinder decomposition to analyse which part of the difference in the NAF of OAMEs and that of establishments can be explained by their enterprise characteristics and what part may be attributed to other factors. We decompose the observed difference in the NAF into two parts—(i) that which is attributed to differences in the enterprise characteristics of OAMEs and establishments, referred to as the ‘endowment’ or the ‘explained’ component, and (ii) the part that is due to differential effects of these characteristics on the NAF of the two different types of enterprises, referred to as the ‘returns to the endowment’ or the ‘unexplained’ component. While the unexplained component is often used as a measure for ‘discrimination’ in literature, it also subsumes the effects of unobserved predictors and therefore should be interpreted as such. The ‘endowment’ component of this decomposition measures how much the NAF of OAMEs would change if they were to have the average characteristics of establishments, while the ‘returns’ component measures how much the NAF of OAMEs would change if the returns to these characteristics for OAMEs were the same as the returns for establishments. This decomposition is an accounting exercise, and we cannot make any causal inferences from it.

We find that the ‘endowment’ component accounts for around 54% of the difference in the NAF between OAMEs and establishments [calculated by dividing the explained part of the difference in NAF (28,227) by the total difference in NAF (51,797), as reported in Model Specification 2 of Table 7]. This component is due to the differences in their enterprise characteristics outlined above, as well as other time-, state- and industry-specific differences. The rest of the difference in the NAF, i.e. 46%, can be attributed to the ‘returns’ component, which cannot be explained by the differences in these enterprise characteristics. This difference can be accounted for, along with other unobserved factors, by the fact that establishments have higher returns to same characteristics (except for subcontracting linkages) than OAMEs, as is evident from the detailed ‘decomposition panel’ in Table 7. We find from the decomposition panel that, not only do establishments have more favourable enterprise characteristics (such as being located outside the household, not being female-headed, being registered etc.) that further their accumulation possibilities vis-à-vis OAMEs, but they also have higher returns to these characteristics relative to OAMEs. However, for certain variables, for example value of assets, the explained and unexplained components of the decomposition have opposite signs. It is possible to explain these opposing signs for the value of assets as follows: the positive sign for the explained component implies that if, on an average, the OAMEs had the same value of assets held as the establishments, their NAF would have been higher than what it was; on the other hand, the negative sign for the unexplained component (though the coefficient is not significant)



Table 7 Oaxaca–Blinder decomposition (dependent variable: NAF)

Differential	Specification 1		Specification 2	
Prediction establishment (1)	60,593.9*** (5071.0)		60,593.9*** (6218.4)	
Prediction OAME (2)	8796.7*** (626.9)		8796.7*** (724.3)	
Difference	51,797.2*** (4689.9)		51,797.2*** (5736.4)	
Decomposition	Explained	Unexplained	Explained	Unexplained
Total	27,564.8*** (2935.6)	24,232.5*** (2695.8)	28,227.2*** (3092.8)	23,570.0*** (3510.1)
Decomposition (detail)				
Time dummy 2005–2006 (base category: year 2000–2001)	−2.183 (8.553)	2309.7 (1629.1)	−1.391 (13.66)	1078.3 (2373.8)
Time dummy 2011–2012 (base category: year 2000–2001)	128.2** (60.51)	1813.7* (1078.0)	157.7 (110.4)	−262.5 (4936.4)
Urban (base category: rural)	1073.8*** (282.3)	2223.2 (2321.4)	870.6* (490.1)	988.2 (2571.7)
Value of assets held (in INR)	12,570.7*** (2300.0)	−226.2 (594.4)	12,195.3*** (2279.1)	−622.7 (605.1)
Contract dummy (base category: non-subcontracted)	156.0* (93.83)	−2739.6** (1132.5)	143.7 (111.0)	−2540.0* (1288.6)
Non-female-headed enterprise (base category: female-headed enterprise)	1683.7*** (251.6)	8074.8* (4162.5)	1488.7*** (200.6)	5172.4 (3938.5)
Located outside household (base category: located within HH)	1526.0*** (308.2)	4301.1*** (1116.4)	1059.5*** (230.5)	5165.3*** (1613.0)
Enterprise registration (base category: unregistered)	4731.8*** (754.2)	2840.9*** (816.1)	4492.4*** (723.9)	2280.5** (970.9)
Accounts maintained by enterprise (base category: accounts not maintained)	5616.6*** (1299.0)	5148.1*** (1371.5)	5494.9*** (1282.6)	4628.7*** (1371.2)
Mixed activity (base category: enterprise does not undertake mixed activity)	70.87* (39.14)	−63.40 (211.7)	67.06* (36.43)	−48.49 (219.1)
Nature of operation: perennial (base category: seasonal/casual)	9.253 (77.24)	13,089.0*** (2348.8)	10.12 (84.52)	26,054.0*** (4661.7)
Constant	−12,538.9*** (3209.8)		35,262.8 (74,620.8)	
Observations	382,993	382,993	382,993	382,993

Robust standard errors clustered at state level in parenthesis; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Specification 1 does not control on state or industry controls [set (b) of controls], whereas Specification 2 controls for them



implies that if, on an average, the OAMEs had similar returns to the value of assets as the establishments, their NAF would have been lower than what it was. In other words, the negative value implies that the OAMEs are able to better utilise their assets, or have higher returns to their assets, than the establishments.¹⁹

While the reason for these differential returns to similar characteristics cannot be observed, a part of them may be attributed, along with other unobserved factors, to the fact that these enterprises encompass different structures and production processes, as indicated in the initial discussion on the dichotomy between OAMEs ('traditional'/non-capitalist enterprises) and establishments ('modern'/capitalist enterprises).

While, with appropriate policy interventions, the accumulation possibilities of OAMEs might be enhanced by improving their enterprise characteristics, e.g. improving credit facilities, locating enterprises outside households, providing 'better' subcontracting linkages with bigger firms that are beneficial to the subcontracted firms etc., some of these characteristics, e.g. the gender of the head of household, being located in rural or urban areas etc., are structurally given and cannot be easily altered by policy interventions.²⁰ Further, there is a significantly large difference in the accumulation possibilities that cannot be explained by these observed characteristics. A part of this unexplained difference in the accumulation possibilities of OAMEs and establishments, as argued above, might be due to an intrinsic difference in their structures and production processes.

The analysis in this and the previous subsection show that the dualism between the (non-capitalist) OAMEs and (capitalist) establishments was reinforced over the considered decade of high economic growth.

Pseudo-panel: a Cohort-Level Analysis

This section presents a panel fixed-effects estimation to control for any additive and time-invariant unobserved heterogeneity that may be correlated with the regressors in the models estimated above. Such unobserved heterogeneity may lead to an omitted variable bias. Since the NSSO does not provide panel data for the unorganised manufacturing sector, here we construct a 'pseudo-panel' to do the analysis.

The pseudo-panel is formed by transforming the available NSSO enterprise-level data into cohort levels. A cohort for a pseudo-panel is defined as "a group with fixed membership, individuals of which can be identified as they show up in surveys"

¹⁹ A similar interpretation can be made for the opposing signs of the explained and unexplained components of the contract dummy. However, the total difference (explained and unexplained) in the NAF between establishments and OAMEs due to subcontracting is negative. As noted herein (and also argued in Basole et al. 2015; Bhattacharya et al. 2013; Raj and Sen 2016), subcontracted enterprises are less productive than non-subcontracted ones, implying that being subcontracted is, overall, an unfavourable characteristic for an enterprise. However, not all values of the decomposition results for the contract dummy are significant.

²⁰ Other structural factors such as caste and religion of the enterprise owner may also affect the economic dynamics of enterprise. However, we are unable to control for these due to unavailability of data for all three rounds of our analysis.



(Deaton 1985, p. 109).²¹ One possible strategy for constructing a cohort from the enterprise-level data is to define cohorts of enterprises over a set of relatively stable characteristics, which allows us to identify particular cohorts in consecutive survey rounds. Here, we define cohort membership in terms of three such characteristics, viz. the state, sector and industry to which the enterprises belong.²² That is, each cohort consists of a set of enterprises belonging to a particular state (we consider 22 major states in India), sector (rural or urban) and industry group (for seven major NICs).²³ The pseudo-panel is then created by ‘collapsing’ each cross-section of data cohort-wise, with the cohort being our new unit of analysis that can be tracked over time.

We estimate the following model:

$$y_{ct} = \alpha_{ct} + \beta X_{ct} + u_{ct}, \quad (3)$$

where $c=1, 2, \dots, n$ indexes the cohorts, y_{ct} is the average NAF for a particular cohort, α_{ct} is the cohort fixed effect, X_{ct} is a vector of cohort characteristics (explained below), β is a vector of coefficients, and u_{ct} is the error term.

The pseudo-panel has 915 observations, resulting in a balanced panel of 305 cohorts at each time period (i.e. 2000–2001, 2005–2006 and 2010–2011).²⁴ Since we are aggregating over cohorts, instead of the dummy variables that we had in our previous analysis, the variables in X_{ct} are the proportions of enterprises within a cohort exhibiting the characteristics of non-base categories. For example, earlier the non-base category for the contract dummy was the enterprise being subcontracted, while here this variable is denoted as the proportion of subcontracted enterprises in a cohort.

One important condition for appropriate application of the fixed-effects model is to have sufficient variation over time within each cohort. We find that all variables, except ‘location’, i.e. the proportion of enterprises located outside the household, have sufficient variation over time, and therefore we estimate this model by including all enterprise characteristics except location (Specification 1 of Model 1). In Specification 2 of Model 1, we drop the variable with the next-lowest within-cohort variation, i.e. the proportion of registered enterprises. Finally, a first difference model (Model 2) with the same set of regressors as in Specification 2 of Model 1 is estimated.²⁵ The results of all three estimations are reported in Table 8.

²¹ Refer to Deaton (1985) and Verbeek (1996) for detailed discussions on the construction of pseudo-panels.

²² Cohorts based on these characteristics do not violate the stratification strategy employed by the NSSO in the surveys.

²³ Among the 35 states and union territories in India in 2011, we include all except Chandigarh, Sikkim, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya, Daman and Diu, Dadra and Nagar Haveli, Goa, Lakshadweep, Puducherry, and Andaman and Nicobar. Of the eight major NICs mentioned earlier, we exclude the tobacco industry (NIC 16) due to reasons outlined in footnote 13 above.

²⁴ Each cohort that we include in the analysis has at least 100 observations, which is sufficiently large to avoid any possible measurement errors (Verbeek 1996; Deaton 1985).

²⁵ The loss in statistical significance for some variables may be due to loss of variation in errors on account of aggregation of a fairly large set of data.



Table 8 Cohort-level pseudo-panel estimation (dependent variable: average NAF per cohort)

Variables	Model 1, Specification 1	Model 1, Specification 2	Model 2
Proportion of establishments	48,187.54** (17,668.07)	67,798.26*** (15,337.39)	65,727.54** (23,640.15)
Total assets	0.01275 (0.01465)	0.01709 (0.01651)	0.02606 (0.02443)
Proportion of subcontracted enterprises	−12,826.95 (15,173.28)	−23,561.90** (11,266.34)	−13,097.90 (15,971)
Proportion of non-female-headed enterprises	2894.17 (13,098.49)	12,768.67** (6688.12)	10,381.51 (10,479.12)
Proportion of enterprises maintaining accounts	50,550.15 (43,645.11)	58,988.76 (58,069.94)	62,228.12 (66,943.47)
Proportion of registered enterprises	85,789.71 (72,926.3)	NA	NA
Proportion of enterprises engaged in mixed activity	774.56 (15,570.7)	11,837.85 (17,415.13)	31,062.14* (16,392.26)
Proportion of perennial enterprises	17,667.20 (19,017.32)	28,229.73 (21,401.52)	29,495.53 (30,854.21)
Constant	−21,162.59 (21,446.5)	−29,569.93 (24,890)	−2668.30 (3170.64)
Time constant for time period 3	NA	NA	7648.23* (6291.51)

Dependent variable: average net accumulation fund (NAF) for a cohort. Number of observations: 915 (305 cohorts for each time period)

Robust standard errors clustered at state level in parenthesis; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

We find that the variable ‘proportion of establishments’ remained highly significant in all specifications, indicating that an increase in the proportion of establishments in a cohort significantly increases its ability to retain a higher NAF, all else being equal. Also, on an average, the proportion of establishments in a cohort remained more or less constant over the decade at around 24%. Further, even at cohort level, we find similar results as above regarding the evolution of the NAF with ‘favourable’/‘unfavourable’ characteristics.

This reaffirms at cohort level—and controlling for unobserved time-invariant heterogeneity—that there was a divergence in the accumulation possibilities of OAMEs (‘traditional’/non-capitalist enterprises) and establishments (‘modern’/capitalist enterprises) over time. Thus, while acting as an additional robustness check for our findings, this analysis provides further credence to the argument about the entrenchment of dualism in the IMS during the period of economic growth.



Conclusions

In literature on development and modernisation, it has been expected that, with economic growth and a consequent increase in per capita income, the dualism between ‘traditional’ and ‘modern’ economic spaces will wither away, leading to a structural transformation and formalisation of the economy. However, in India, in spite of a long period of sustained economic growth, the ‘traditional’ informal sector has continued to persist and to provide employment to the vast majority of the population. Existing literature does not analyse the evolution of the informal sector in India during the period of high economic growth, while linking it to the broader process of structural transformation. This article is an intervention in this respect.

We identify an ingrained dualism between the ‘traditional’/non-capitalist segment and ‘modern’/capitalist segment in the IMS and find that there has not been a marked tendency towards a transformation of the IMS even during the period of high growth. One possible channel for this transformation is through the process of ‘traditional’ enterprises transitioning into larger, ‘modern’ ones. However, the OAMEs (the ‘traditional’/non-capitalist enterprises), which comprise the vast majority of IMS enterprises, were not able to retain any substantial fund for accumulation (NAF) during this period. Moreover, there was an increasing divergence in the accumulation capabilities of these OAMEs and the establishments (the ‘modern’/capitalist enterprises)—a divergence that is further underscored by their different enterprise characteristics. Not only did the dualism between the ‘traditional’ and ‘modern’ segments of the IMS remain intact, but it seems that it was reinforced and became further entrenched during the high-growth decade. While there exists the possibility that *some* ‘traditional’ enterprises may be able to grow into establishments over time given a rise in their average NAF, our analysis finds that overall there has not been any marked tendency towards a dilution of the structure of economic dualism in the IMS.

Another possible channel for this transformation is that ‘traditional’ enterprises may be unable to economically reproduce themselves over time and eventually wither away. However, we find that, on an average, the (‘traditional’) OAMEs have been able to retain a positive, though trivial, amount of NAF and thereby are able to reproduce themselves over time, even though at a very low level.

The persistence of dualism and a possible stalling of the transformation process in many labour-surplus developing economies have also been raised in some recent literature (Binswanger-Mkhize 2013; De Vries et al. 2012; La Porta and Shleifer 2014; Timmer and Akkus 2008). The question that remains relevant, in the Indian context, is why these ‘traditional’ enterprises continue to operate and reproduce themselves even at such precariously low levels. A possible explanation is provided by a recent strand of literature which attributes this reproduction to the ‘exclusionary’ nature of the growth and development process in India (see for example Bhaduri 2017; Bhattacharya 2017; Bhattacharya and Kesar 2018; Chakrabarti 2016; Chen 2014; Harriss-White 2012; Sanyal 2007; Sanyal and Bhattacharyya 2009). It argues that the growth process transfers resources from



the ‘traditional’ economy to the growth-oriented ‘modern’ sector, without providing formal employment to those who lose their livelihoods due to such transfers.

In particular, following Sanyal (2007), it has been argued that the vast ‘excluded’ population that is not absorbed in the ‘modern’/capitalist sector driving the process of economic growth can survive only by *reproducing* the ‘traditional’/non-capitalist economic spaces—even though precariously on a thin resource base—that provide their means of livelihood. The continued persistence of the ‘traditional’ IMS enterprises, even when they are performing precariously, can therefore be partly explained in terms of the overarching principle of satisfying the subsistence needs of the households that might govern the economic logic of reproduction of these enterprises. Further, the overlap of the household space of consumption and the enterprise space of production might allow these firms to “commonize” or to draw from various household resources, such as physical space of the household, family labour, household tools etc., and thus help them to reduce explicit production costs, thereby helping them to survive (Bhattacharya et al. 2013). The presence of such overlaps also allows these household enterprises to draw resources from other household sources or to divert the fund for accumulation to augment household consumption, when needed (Bhattacharya 2017). This process of recreating the ‘traditional’ informal spaces for survival of the vast segment of population thereby ensures the continued reproduction of dualism—a phenomenon that is antithetical to the arguments about economic transformation that animate much of the debates in development literature.

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Compliance with Ethical Standards

Conflict of Interest On behalf of all authors, the corresponding author states that there are no conflicts of interest.

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